PRAJWAL A S

Python Backend Developer · AI Agents & RAG Systems

Bangalore, India | prajwalas8999@gmail.com | github.com/prajwalas1 | linkedin.com/in/prajwal-a-s

PROFESSIONAL SUMMARY

Python backend developer with experience building GenAI systems, REST APIs, and LLM-powered AI agents. Co-designed and deployed an agentless telemetry pipeline (using existing Redfish API and SSH protocols, no custom agent software) and an NL-to-SQL AI assistant on a 422-node petascale supercomputer, with work submitted to SC26 (Supercomputing 2026, under review). Built a real-time, multilingual voice AI agent using a tool-calling architecture so the LLM never invents data — every action is grounded in a real backend call. Comfortable with async Python, RAG pipelines, prompt engineering, and PostgreSQL schema design, and experienced delivering systems reliably in secure, restricted production environments.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Backend: FastAPI, REST APIs, Async Python (asyncio), Pydantic, Uvicorn

GenAI / LLM: LLMs, AI Agents, RAG Pipelines, Prompt Engineering, LangChain, LangGraph, Ollama, Embeddings, pgvector, Speech-to-Text, Text-to-Speech, Voice AI

Databases: PostgreSQL, pgvector

AI Infra: Self-hosted GPU Inference (H100), Ollama, LiveKit, Voice AI, Vector Databases, Token Optimization

Monitoring: Grafana, Loki, Promtail, Webhooks, Alerting Pipelines

Systems: Linux, Git, SSH, Redfish API, InfiniBand, iDRAC/BMC, Paramiko

Frontend: React, Grafana Embedded Widgets, Custom HTML Panels

EXPERIENCE

Project Associate - I | CSIR Fourth Paradigm Institute (CSIR-4PI)

Sep 2025 – Mar 2026 | Bangalore, India | HPC Infrastructure & AI Systems

- Architected a production NL-to-SQL AI agent enabling cluster operators to query live HPC telemetry in natural language, achieving 93.0% accuracy across 1,000 production queries with a locally-deployed LLM.
- Designed a 7-stage async FastAPI pipeline: intent classification → schema grounding → few-shot prompt construction → LLM SQL generation → validation → PostgreSQL execution → NL response synthesis.
- Built a really fast LLM inference pipeline through prompt restructuring, token reduction, and RAG-style few-shot context injection from a 500 query-SQL pair repository.
- Built and benchmarked multiple open-source LLMs (qwen2.5:7b, mistral:7b, codellama, gpt-oss:20b) via Ollama on H100 GPUs, selecting the optimal model on accuracy and speed.
- Built automated telemetry ingestion pipelines using existing SSH (Paramiko) and Redfish API protocols — requiring no custom agent installation — collecting 38 metrics across CPU, GPU, memory, network, power, and storage subsystems from 422 nodes into PostgreSQL, accumulating 132 million records.
- Designed a unified 8-table PostgreSQL schema co-locating hardware telemetry with job scheduler metadata, enabling cross-domain queries in 151ms on 36M+ row tables.
- Built RAG-style schema compression, cutting inference cost by ~60% while improving SQL generation accuracy.
- Implemented a multi-level proactive alerting pipeline with a Flask-based webhook relay for threshold-triggered email alerts in a fully air-gapped, network-restricted environment.
- Built Grafana dashboards to visualize what's happening across the HPC cluster in real time, at cluster, rack, node, and GPU-card levels.

RESEARCH (UNDER REVIEW)

SC26 Submission — Supercomputing 2026 (Under Review), Apr 2026

"An Agentless Telemetry Framework with Job-Correlated Proactive Alerting and Natural Language Querying for Air-Gapped HPC Systems," CSIR Fourth Paradigm Institute

- Dual-protocol agentless telemetry (Redfish API + SSH) collecting 38 metrics from 422 HPC nodes without agent installation.
- NL-to-SQL interface with a locally-deployed LLM achieving 93.0% accuracy on 1,000 production queries in an air-gapped environment.

KEY PROJECTS

AI Voice Appointment Assistant — Healthcare Voice AI Agent
Python · FastAPI · LiveKit Agents · React · PostgreSQL (Neon) · Sarvam AI (STT/TTS)
Live Demo | GitHub Repository

- Built a real-time, multilingual (English, Hindi, Kannada) voice AI agent that lets patients book, cancel, reschedule, and check appointments over WebRTC, using LiveKit Agents for the streamed speech-to-text → LLM → text-to-speech pipeline.
- Designed a tool-calling architecture so the LLM never invents appointment IDs or availability — every action is backed by a real FastAPI + PostgreSQL database call, keeping outputs grounded and auditable.
- Implemented returning-patient recognition via phone-number lookup and built the REST API layer (FastAPI, SQLAlchemy) powering the agent, deployed across Vercel, Render, and LiveKit Cloud.

AI-Powered HPC Monitoring Assistant — CHAMP Supercomputer
Python · FastAPI · PostgreSQL · pgvector · LangChain · LangGraph · Ollama · React · Grafana

- Built a multi-agent prompt pipeline with deterministic, multi-stage reasoning, eliminating hallucinated tool calls and making failure modes explicit and debuggable.
- Implemented domain-specific SQL generation rules for GPU node filtering, time-bucketing, and job-count queries across 8 telemetry tables, deployed daily on live HPC infrastructure.

EDUCATION

B.E. — Computer Science & Engineering
Malnad College of Engineering, Hassan · Visvesvaraya Technological University (VTU) · Aug 2019 – Nov 2023

KEY ACHIEVEMENTS & METRICS

- 93.0% NL-to-SQL accuracy on 1,000 real HPC production queries using a locally-deployed LLM.
- 151ms query execution for a cross-domain join across 36M+ row PostgreSQL tables.
- Built a really fast LLM inference pipeline through prompt engineering and token optimization.
- 60% inference cost reduction via RAG-style token-efficient prompt compression.
- SC26 research paper submission (under review) — one of the most competitive HPC venues globally.